

Atul Gupta

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Summary

Dynamic AIML Engineer with 5 years of experience in advanced data analysis, data modeling, and cloud engineering. End-to-end product builder with deep interest in deep learning, natural language processing, artificial intelligence, and business process automation. 3x AWS certified.

Education

Master of Science in Computer Science (GPA: 3.96): Stevens Institute of Technology, Hoboken, NJ

Sep 2022 – May 2024

- **Courses:** Machine Learning, Deep Learning, Data Mining with R, Big Data, Agile development, Algorithms
- **Tech:** Sklearn, PySpark, MongoDB, Javascript, Bootstrap, Java, Python

Bachelor of Technology in Electrical and Electronics Engineering: VIT University, Vellore, India

Jun 2013 – Jun 2017

Technical Skills

Programming languages: Python, SQL

Frameworks: Pytorch, Scikit-learn, Tensorflow, transformers, NLTK, NumPy, Pandas, Flask, PySpark

Cloud: AWS, Sagemaker, Bedrock, Pinecone

Tools: Jupyter, Service Now, Docker, Kubernetes, Github, Airflow, MS SQL Server, LLM, Powershell

Experience

Application Development Analyst (Data Analyst)

Jan 2021 – Jul 2022

Accenture

- Engineered cloud-native ELT pipelines using airflow and glue to analyze transactional and guest behavior data from Redshift.
- **COVID-19 revenue forecasting model (Tech used: Python, SQL Server, PyTorch, XGBoost, Docker, AWS Sagemaker):** Created ensemble machine learning model that predicted guest reservations and revenue for top 50 Hilton properties during the COVID disruption with mean absolute percentage error of 19%. Integrated with SSRS to deliver daily forecast reports to top hotels.
- **Guest churn prediction model (Tech used: Python, Sklearn, SQL Server, AWS Sagemaker):** Modeled 2 years of historical data from Hilton Loyalty Program, Hilton Honors, to create a classifier that identifies churn-prone members with 85% accuracy enabling targeted marketing campaigns increasing loyalty by 10%.
- **Enhancing patch deployment application (Tech used: Lambda, Cloudwatch, SNS):** Evolved the patch deployment application by implementing new features to check for pre-requisites, remediate known failures, create detailed reports on installation, and notify concerned teams via emails thereby reducing deployment failures by 90%.

Application Development Associate (Database administrator)

Jan 2019 – Dec 2020

Accenture

- **Automation and six-sigma:** Led and implemented 7 Six Sigma Lean and Yellow Belt projects, automating 15 business processes saving 10k man-hours worth of effort annually and \$30k for the client in third-party service fees. Recognized as **TechStar** by Accenture.
- **Patch deployment automation (Tech stack: Python, Service Now, Powershell):** Engineered and deployed an automated patch deployment application for Hilton's OnQ PMS across 6,000+ Windows servers worldwide to reduce manual effort by 80%, down-times by 90%.
- Resolved 150+ unique critical issues by understanding enterprise application architecture in-depth, conducted knowledge transfer sessions with a team of 30, and presented root cause analysis to client executives.
- **Database administration:** Managed 6000 production database instances worldwide ensuring consistency, reliability, and durability of the data. Created new SQL stored procedures and jobs to ensure compliance with existing rules and regulations globally.

Software Developer

Jun 2017 – Dec 2018

Tech9logy Creators (Startup)

- Prototyped the PoCs on serverless AWS architecture (S3, Lambda, DynamoDB) for test runs and presented the findings to the clients.

Projects

RAG AI Personal Assistant | https://github.com/atulgupta002/Personal_Assistant_RAG_AI

- A custom AI NLP system that facilitates hiring professionals to learn interactively about a candidate's profile.
- **Domains:** natural language processing, retrieval augmented generation, vector search, prompt engineering, guardrails.
- **Tools:** sentence transformers, FAISS, Python, Scikit-learn, Flask, AWS Elastic Beanstalk, Bedrock, EC2, S3, Linux, Docker, LLM

Fine-tuned BERT | https://huggingface.co/atulgupta002/banking_customer_service_query_intent_classifier

- A custom, fine-tuned BERT transformer to classify customer service queries into different intents trained on synthetic dataset.
- **Tools:** Google BERT-base-uncased, hugging face transformers, PyTorch, Python, NLTK, CUDA
- **Results:** Observed evaluation loss of 0.06 on test dataset after training for 3 epochs.

Certifications

AWS Certified Machine Learning Engineer Associate ([Link](#))

AWS Certified Solutions Architect Associate ([Link](#))

AWS Certified AI Practitioner ([Link](#))